

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 15

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BIOTECHNOLOGY

CRISPR Cas 9

- It is a gene editing technique that can target and modify DNA with a lot of accuracy.
- The system consists of CRISPR, Cas and guide RNA.
- Scientists in Japan were the first to discover CRISPR in the DNA of bacteria in 1987. The scientists noticed a pattern of short, repeating, palindromic DNA sequences separated by short non-repeating 'spacer' DNA sequences.
- Later scientists discovered that this pattern is present in many bacteria.
- In 2012 the term CRISPR was coined which stands for Clustered Regularly Interspaced Short Palindromic Repeats.
- Cas9 (CRISPR-associated protein 9) is an enzyme that uses CRISPR sequences as a guide to recognize and cut specific strands of DNA that are similar to the CRISPR sequence.



BIOTECHNOLOGY

CRISPR Cas 9

- Scientists found out that the entire system (CRISPR, Cas & RNA) plays a role in the bacterial immune system.
- When a bacteria encounters an invading DNA such as from a virus, the bacteria can copy and incorporate segments of foreign DNA into their genome as 'spacers' between the short DNA repeats.
- These spacers enhance the bacteria's immune system by enabling RNA molecules to quickly identify and target the same DNA sequence in the event of future viral invasion.
- If the RNA molecules recognize an incoming sequence of foreign DNA, they call the Cas proteins which are specialized for cutting DNA. They cut and disable the invading genome.
- Cas9 is an RNA-guided enzyme that cuts foreign DNA bearing sequence similar to the RNA loaded into the enzyme during bacterial adaptive immunity.

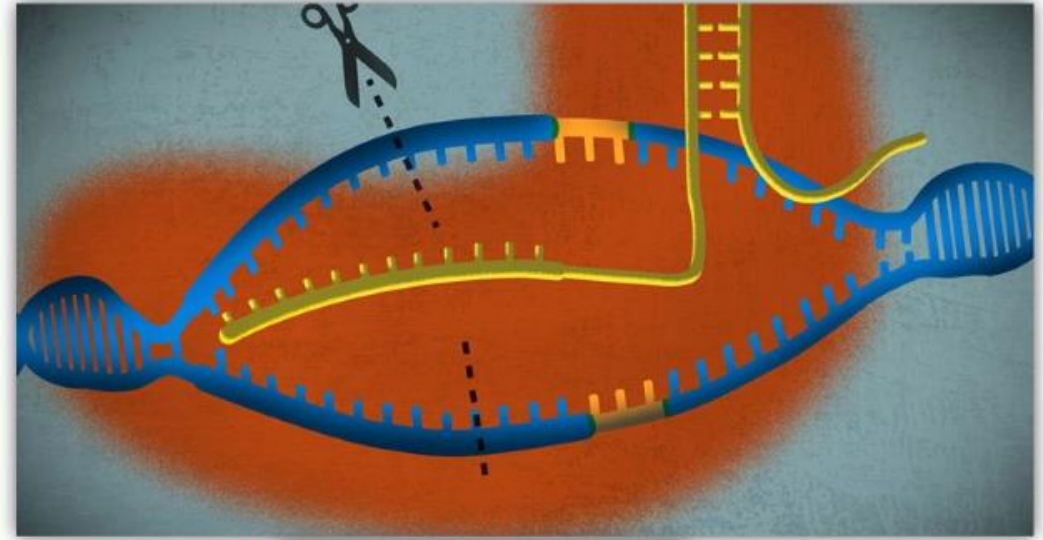
MECHANISM

- An enzyme called Cas9 acts as a pair of 'molecular scissors' that can cut the two strands of DNA at a specific location in the genome so that pieces of DNA can then be added or removed.
- A piece of RNA called guide RNA (gRNA) guides Cas9 to the right part of the genome. This makes sure that the Cas9 enzyme cuts at the right point in the genome.
- The Cas9 follows the guide RNA to the same location in the DNA sequence and makes a cut across both strands of the DNA.



MECHANISM

- In 2012, scientists from the US used this system to create a new gene editing tool that could identify, cut and replace any gene sequence.
- The main goal of scientists working in labs is to isolate and manipulate genes. It is easy to isolate a sample of DNA from a group of cells. But finding a specific gene within DNA can be compared to finding a needle in a haystack.
- Each human cell contains 6 feet of DNA, so a small tissue sample contains several kilometres long DNA.
- The CRISPR Cas 9 technology has made it relatively easy to isolate a gene, enabling researchers to determine its nucleotide sequence, study its genetic data and manipulate it.



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